ChatGPT

Designing an Explainable Multimodal Engagement Model

A robust engagement model should fuse **facial cues** (real-time webcam) with **learning analytics** (quiz scores, time-on-task, LMS logs, etc.). In practice this means extracting *interpret-able features* from video (e.g. facial action units, gaze direction, head pose) and behavioral logs, then combining them in a transparent model. For example, open-source tools like OpenFace or MediaPipe can extract **facial Action Units (AUs)** and gaze; behavioral data includes quiz responses, clickstreams, time on pages, tab switches, mouse/keyboard activity and forum posts. Recent work shows that such multimodal fusion dramatically improves accuracy: Das & Dev found that adding AUs to an XGBoost model yielded **82.9%** accuracy on the DAiSEE engagement dataset (versus only ~47% for a CNN alone) [1]. Similarly, Xie *et al.* fused face expressions (VGG16), head-pose (ResNet) and facial landmarks, achieving ~62% accuracy and strong correlation (r=0.714) with self-reported engagement [2]. In short, **combine many signals**: use CNN or ML to process video into features (smiles, nods, blinking, gaze fixation) and merge with LMS metrics (quiz scores, time spent, pages accessed) into one model [1] [3]. A typical architecture (Fig.1) preprocesses each modality (audio/video normalization, log filtering), extracts features (CNN for faces, embedding for text/logs), then fuses them (via concatenation, attention, or ensemble models) into an engagement classifier [3] [1].

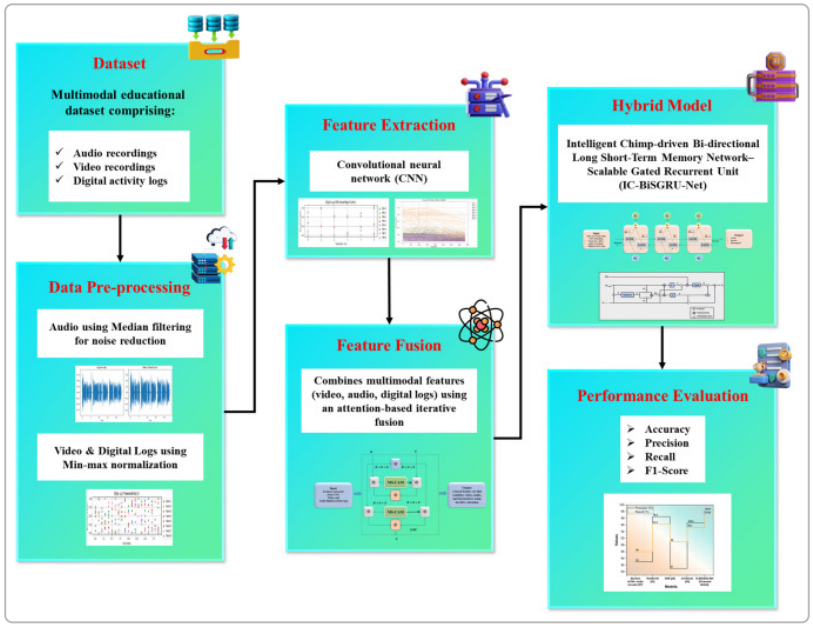

Figure: Example pipeline for multimodal engagement detection (audio/video + logs) [1] [3]. The system in Fig.1 shows how video frames and log data are normalized (filters, scaling), features (facial AUs, micro-expressions, gaze, log metrics) are extracted via a CNN, and then a hybrid model (e.g. LSTM or GRU) fuses these for engagement classification.

Key tip: Use **interpretable** features and models. For example, instead of an opaque deep network, one might feed the extracted AUs, gaze angles, quiz score, and page-view counts into a decision tree, random forest or an XGBoost model. Feature-importance analysis (e.g. SHAP) can then reveal which cues most drive engagement [4] [5]. Zhao *et al.* (2024) demonstrate that fuzzy-logic models can capture uncertainty in educational data while yielding human-readable rules [4]. One could similarly build a **fuzzy inference system** that says e.g. "IF eye-contact is low AND quiz score is low THEN engagement = low", making the scoring transparent.

Improving Accuracy and Model Design

The current DAiSEE-based model (60% acc.) can be improved by richer data and better algorithms. **Augment data and features:** include more modalities and diverse examples. In addition to face AUs, track **eye openness, smile intensity, head tilt**, etc. Augment the training set via data augmentation (synthetic head rotations, lighting changes) or by collecting new video from your own LMS. Use external datasets (EmotiW student engagement, WACV16) for pretraining. Ensure classes are balanced or use weighted loss (DAiSEE is imbalanced).

Explore model choices: Recent studies suggest classical ML can outperform deep nets on engagement when well-engineered. For instance, Das & Dev found that a gradient-boost (XGBoost) on AU+behavioral features far outperformed a deep net [1]. Try models like XGBoost, Random Forest, SVM on extracted features; also experiment with deep learning (CNN+LSTM, Temporal Convolutional Networks, attention). Hybrid architectures (e.g. CNN for faces + LSTM for temporal context) can capture dynamics. For example, Hou *et al.* (2023) used a BiLSTM+SGRU network on video+audio+logs and reported 96–99% accuracy on three engagement states [6].

Cross-validate and ensemble: Compare single-modality vs. multimodal performance. Use cross-validation to avoid overfitting. An ensemble of a deep video model and a tree-based model on features (like "dual-path" fusion) can further boost accuracy. Liu *et al.* (2024) found that blending CNN frames with traditional classifiers yielded ~68% on DAiSEE, slightly better than any one method [7].

Finally, consider **temporal smoothing**: raw frame-by-frame scores are noisy. Aggregate over sliding windows (e.g. 5–10 sec) to get a stable score. Report both momentary and session-level engagement.

Interpretability and No "Black Box"

Since the goal is a **trustworthy, explainable** system, avoid inscrutable models. Use interpretable ML or augment models with explainability tools: for instance, run a tree-ensemble and then use SHAP to highlight which features (smile vs. quiz score vs. tab-switch frequency) drove each prediction [4] [5]. Cheng *et al.* (2023) even proposed a "gradient magnitude mapping" to quantify each feature's contribution to the engagement score [5]. These techniques let a teacher understand *why* a student was scored as "disengaged" (e.g. "80% due to low attention and 20% due to incorrect answers").

Using **fuzzy logic** is another strategy: Zhao *et al.* note that fuzzy systems naturally handle uncertainty and yield human-like reasoning rules [4]. A fuzzy rule set could classify engagement (high/medium/low) based on linguistic inputs (e.g. "eye contact: high/medium/low; quiz performance: good/fair/poor"). Such a rule-based model is transparent by design.

Engagement scores should be explained to teachers. An "open learner model" interface could display, for each student, the main signals affecting their score (smiling, nodding, answering questions). Khosravi *et al.* emphasize that **transparency builds trust** in educational AI [8]. Indeed, if teachers can see which behaviors the model used, they can better act on or dispute the results.

Real-time vs. Post-lecture Analysis

Real-time monitoring: In live sessions, run the webcam-based model continuously and update an "engagement meter" or attention chart. For speed, you may simplify features (e.g. track only eyes and mouth openness) or use a lightweight model. Smooth the score over seconds to reduce flicker. Provide teachers with a live dashboard: e.g. "X% of students appear attentive now."

Post-lecture analytics: After a class, use collected data to generate insights and reports. For example:
- **Engagement heatmaps:** Map time in the video or lecture to aggregate attention. Learning analytics tools often visualize "watched-time" as a heatmap [9]. High-intensity segments (red) show where students replayed or lingered, low (blue) where many skipped. This pinpoints confusing sections. For instance, if a major drop in attention coincides with a complex topic, the teacher can revisit that part in the next session.
- **Topic difficulty mapping:** Correlate quiz performance with segments of the lecture. A heatmap over slides or topics can highlight "pain points" where scores dipped. Combine this with log data: e.g., many students pausing videos to rewind suggests struggle.
- **Individual vs. cohort views:** Provide each teacher with a summary: how each student's engagement evolved (charts of face-score vs time) and a group view (distribution of engagement scores over time). This helps tailor follow-ups (e.g. give supplementary materials on parts where class-wide engagement was low).

For example, universities often show instructors bar charts of forum contributions or video interactions by student [9]. A dashboard might display, say, how often each student asked questions or annotated notes. Fig.56 (below) illustrates a sample analytics dashboard (here forum participation by student) – similar visualizations can guide interventions.

Figure: Sample engagement dashboard (forum activity by student). Such analytics let teachers see which students are actively contributing versus passively observing.†

Integrating the ICAP Framework

The **ICAP framework** (Chi & Wylie 2014) categorizes learning activities into four modes: Passive, Active, Constructive, and Interactive [10]. Each successive mode (P→A→C→I) is associated with deeper processing and better learning outcomes [10] [11]. For example, simply listening to a lecture is *Passive*, answering a quiz question is *Active*, generating a summary or drawing a concept map is *Constructive*, and engaging in a group discussion is *Interactive*.

To integrate ICAP, label each LMS event by its mode. E.g. **watching lecture video** = Passive; **doing a drill or quiz** = Active; **writing a reflective summary or asking a question** = Constructive; **collaborating on a project/chat** = Interactive. Log these for each student. Then analyze: do students who spend more time on *Constructive/Interactive* tasks show higher quiz scores or final grades? Prior studies suggest they should – one experiment found that students given active preparation (self-study and question-generation) significantly outperformed those given passive (lecture) preparation [12]. Chi's ICAP hypothesis predicts this ordering: "Interactive leads to deeper learning than Constructive, which is higher than Active, which in turn is higher than Passive" [10].

A separate ICAP-focused study could, for instance, compare two teaching designs: a "traditional" class (mostly lectures) versus an "ICAP-oriented" class (with added discussion, peer instruction, and generative tasks). Measure outcomes (exam scores, retention) and engagement metrics. The thesis can test whether the ICAP-oriented class yields significantly better engagement and learning [12] [10]. Write-up should explain ICAP theory (citing Chi & Wylie), detail how student activities were classified, and report quantitative results. Even if an experimental comparison is not possible, one can analyze existing course logs: e.g. cluster student behavior profiles by ICAP-mode frequencies and correlate with outcomes.

Ensuring Publishable Research

To make the work publishable, focus on **novelty, rigor, and insights**. Novelty might come from: (1) combining multimodal engagement sensing *with* a cognitive framework like ICAP, (2) introducing an interpretable scoring algorithm, or (3) providing new analytics (e.g. a real-time dashboard evaluated in classrooms). Thoroughly review recent literature (as above) to highlight gaps your work fills.

Key steps for rigor: collect adequate data (multiple lectures, many students), perform robust evaluation (hold-out test sets, cross-validation), and compare against baselines (e.g. accuracy of simple log-only model vs. your multimodal model). Report not just accuracy but also *explainability*: for instance, present the main decision rules or feature importances that your model uses to assign engagement scores. Discuss ethical aspects: ensure student consent, data privacy, and fairness (e.g. does the model work equally for all genders/ethnicities?). Zhao *et al.* note that fuzzy-rule models can enhance reliability and fairness in educational assessments [4].

For writing, clearly separate the engagement-detection work from the ICAP study, as suggested. The ICAP thesis should focus on educational outcomes and cognitive engagement theory, while the engagement model paper should focus on the sensing/ML method and accuracy. In both, use descriptive headings and bullet points (e.g. list "features used" or "steps in model"). Use the cited sources to justify design choices (e.g. "we extract AUs as in [59†L79-L84]" or "we visualize video heatmaps as recommended by LA research [9]").

Finally, prioritize **interpretability and validation**: as Mathrani *et al.* emphasize, transparent models increase trust and adoption in education [8]. Providing clear explanations for engagement scores (via features or rules) will make the research stand out. Extensive evaluation on real or benchmark data, and releasing code or data (if possible), will strengthen publication potential. In summary, build a multimodal, hybrid model (face + logs), use explainable methods (trees/fuzzy/XAI), integrate ICAP-based analysis separately, and thoroughly validate – this approach has strong support in recent studies [1] [5] [10] and can yield both high accuracy and meaningful insights.

Sources: Recent literature on multimodal engagement detection, explainable educational models, and ICAP theory were used throughout [1] [3] [4] [10]. These include studies on feature fusion (face, logs, etc.), deep learning approaches, fuzzy-logic interpretation, and empirical evaluations of ICAP-based active learning [1] [2] [12] [10]. All recommended practices above are grounded in these peer-reviewed and open-access sources.

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